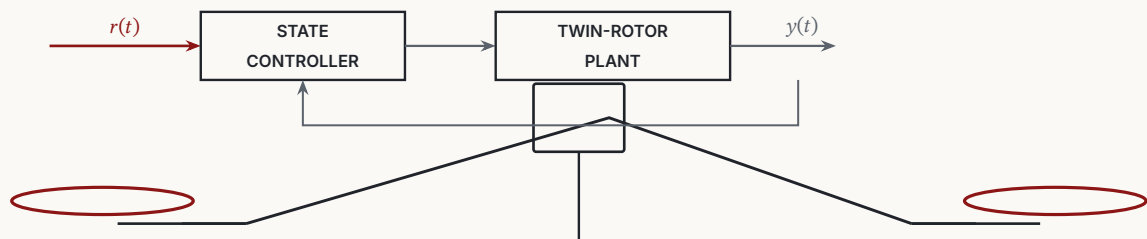


CONTROL SYSTEMS ENGINEERING

Experimental Identification and Modern Control of a Twin-Rotor Helicopter

Integral state feedback and linear-quadratic regulation:
modeling, real-time implementation, and performance comparison



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Abstract

This project consolidates two experimental control studies conducted on the same three-degree-of-freedom twin-rotor helicopter platform. The work is organized as one coherent engineering investigation rather than as two independent laboratory exercises. The platform is first identified in open loop through its elevation, pitch, and yaw channels. The resulting models are assembled into a five-state multivariable representation whose regulated outputs are elevation and yaw rate. Two integral control strategies are then developed and compared.

The first strategy is a discrete integral state-feedback controller designed by pole placement for the elevation channel. A refined second-order model is discretized with a zero-order hold at $T_s = 0.05$ s and augmented with an integral state. The nominal placement, based on a continuous-time design horizon of -1 , gives $K_p = [10.6434 \ 10.2527]$ and $K_i = 0.2882$. It produces a non-oscillatory response with zero steady-state error and a simulated response time of approximately 6 s; the measured response time on the real platform is 5.78 s, with recovery from manual disturbances in approximately 4.41 s. A faster design moves the horizon to -2 and reduces the response time to about 3 s, but increases the peak control demand to approximately 3.3 V.

The second strategy is a multivariable linear-quadratic regulator with integral action. The seven-state augmented model is weighted by $Q = \text{diag}(0, 0, 0, 0, 0, 40, 40)$ and $R = I_2$, with a spectral shift $\alpha_c = 0.5$. The resulting closed-loop poles all lie in the open left half-plane. Simulations report 5% response times of approximately 3.1 s for elevation and 2.7 s for yaw rate, no actuator saturation, and rejection of simultaneous channel disturbances in approximately 5 s. Real-time dSPACE tests confirm stable tracking and disturbance rejection.

The comparison shows that pole placement offers direct control over modal speed and a transparent SISO implementation, whereas LQR provides a systematic multivariable compromise between tracking performance and actuator effort. Both controllers validate the state-space approach, but the LQR architecture is better suited to coordinated control of the coupled helicopter platform.

KEYWORDS twin-rotor helicopter; system identification; discrete state feedback; integral action; pole placement; LQR; Riccati equation; dSPACE; multivariable control.

Project at a Glance



Twin-rotor experimental platform used for system identification, controller synthesis, and real-time validation.

ENGINEERING PROBLEM

Identify the three-axis helicopter dynamics and achieve accurate, stable, disturbance-rejecting control without excessive motor voltage.

SYSTEM MODEL

Elevation: underdamped second order; pitch: double integrator; yaw: integrating first-order dynamics. Combined plant: five states, two motor-voltage inputs, two regulated outputs.

CONTROLLER 1

Discrete integral state feedback by pole placement for elevation. Two speed settings are evaluated and implemented through MATLAB/Simulink and dSPACE.

CONTROLLER 2

Continuous multivariable LQR with integral augmentation, Riccati synthesis, and explicit weighting of accumulated tracking errors and actuator effort.

COMPARATIVE RESULT

Pole placement gives direct transient shaping; LQR coordinates both channels and formalizes the performance-effort trade-off while preserving actuator headroom.

The mathematical models, numerical values, experimental plots, simulation results, and implementation images are retained from the two original reports. The English prose and document structure have been rewritten as a unified research project.

Nomenclature

$\epsilon, \theta, \lambda$	Elevation, pitch, and yaw angles.
$v = \dot{\lambda}$	Yaw angular velocity.
u_g, u_d	Left- and right-motor command voltages.
H_e, H_t, H_l	Elevation, pitch, and yaw transfer functions.
K_e, K_t, K_l	Identified channel gains.
ξ, ω_0	Damping ratio and undamped natural frequency of the elevation mode.
τ_l	Yaw-channel time constant.
A, B, C, D	Continuous-time state-space matrices.
F, G	Discrete-time state and input matrices.
F_e, G_e	Integral-augmented discrete plant matrices.
A_a, B_a	Integral-augmented continuous multivariable matrices.
K_p, K_i	State-feedback and integral gains.
Q, R	LQR state and control weighting matrices.
P	Positive-definite solution of the algebraic Riccati equation.
T_s	Sampling period.
T_c	Pole-placement design time constant.

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Introduction

The twin-rotor helicopter rig is a compact but demanding benchmark for modern control. Its motion combines three physically distinct degrees of freedom: elevation ϵ , pitch θ , and yaw λ . The two electric motors act through their sum and difference, while the measured variables are coupled by the mechanical geometry of the platform. The system therefore provides a useful progression from single-input single-output identification to multivariable state-space control.

The objective of this project is to build one complete control-engineering narrative from two experimental campaigns. The first campaign characterizes the full platform and develops a linear-quadratic controller for simultaneous elevation and yaw-rate regulation. The second campaign focuses on the elevation channel and develops a digital state-feedback controller with integral action. Rather than presenting these studies independently, the report places them in a common framework and compares their design logic, transient performance, actuator demand, disturbance rejection, and real-time implementation.

The project follows four steps. First, open-loop experiments are used to identify transfer functions for elevation, pitch, and yaw. Second, the channel models are assembled into a multivariable state-space description. Third, an integral state-feedback controller is designed in discrete time by pole placement and tested at two response-speed settings. Fourth, a multivariable LQR is synthesized from an augmented continuous model and validated in simulation and on the dSPACE-controlled platform.

The main performance objectives are:

- asymptotic tracking with zero steady-state error;
- stable and well-damped transient behavior;
- limited motor-voltage demand and avoidance of saturation;
- rejection of external mechanical disturbances;
- consistency between simulation and real-time experiments;
- a clear comparison between explicit pole placement and optimal quadratic regulation.

System Modeling and Experimental Identification

DESIGN STAGE 1 Experimental platform and input-output decomposition

1.1 Twin-rotor platform and controlled coordinates

The experimental helicopter can rotate about three axes. Elevation is the vertical rotation of the complete arm, pitch is the local rotation of the rotor cradle, and yaw is the azimuthal motion around the vertical support. Dedicated sensors provide real-time measurements of the relevant angles and angular rates. The plant is actuated by the left and right motor voltages, u_g and u_d .

The sum and difference of the motor commands provide a natural decoupling for identification:

$$u_\Sigma = u_g + u_d, \quad u_\Delta = u_g - u_d.$$

The elevation mode is mainly driven by u_Σ , whereas pitch is mainly driven by u_Δ . Pitch subsequently generates yaw motion through the geometry of the arm.



Figure 1.1: Experimental twin-rotor helicopter platform and representative configurations.

1.2 Elevation-channel identification

DESIGN STAGE 2 Open-loop identification of the underdamped elevation mode

The elevation transfer function is defined by

$$H_e(s) = \frac{\epsilon(s)}{u_g(s) + u_d(s)}.$$

During the experiment, pitch and yaw are mechanically constrained and a voltage step is applied to both mo-

tors. The measured elevation response is oscillatory and is modeled as a second-order underdamped system,

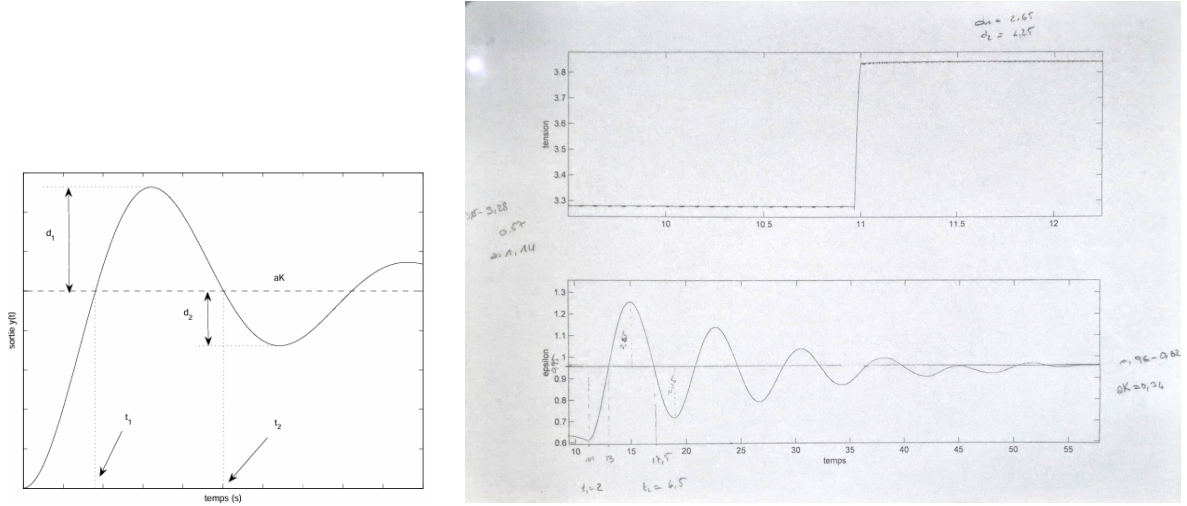
$$H_e(s) = \frac{K_e}{1 + \frac{2\xi}{\omega_0}s + \frac{1}{\omega_0^2}s^2}.$$

For a step of amplitude a , the response is

$$y(t) = aK_e \left[1 - \frac{e^{-\xi\omega_0 t}}{\sqrt{1-\xi^2}} \sin(\omega_0\sqrt{1-\xi^2}t + \phi) \right], \quad \phi = \arctan\left(\frac{\sqrt{1-\xi^2}}{\xi}\right).$$

Successive extrema give

$$\frac{d_2}{d_1} = \exp\left(-\frac{\pi\xi}{\sqrt{1-\xi^2}}\right), \quad \omega_0 = \frac{\pi}{\sqrt{1-\xi^2}} \frac{1}{t_2 - t_1}.$$



(a) Identification quantities for an underdamped response.

(b) Measured motor command and elevation response.

Figure 1.2: Elevation-channel experimental identification.

The measurements reported in the multivariable campaign are

$$aK_e = 0.34, \quad K_e = 0.29,$$

$$d_1 = 2.65 \text{ cm}, \quad d_2 = 2.15 \text{ cm}, \quad \boxed{\xi = 0.05},$$

$$t_1 = 2 \text{ s}, \quad t_2 = 6.5 \text{ s}, \quad \boxed{\omega_0 = 0.7 \text{ rad s}^{-1}}.$$

The very low damping ratio explains the long-lived oscillations observed in open loop.

1.3 Pitch-channel identification

DESIGN STAGE 3 Identification of the differential motor channel

Pitch is driven primarily by the voltage difference,

$$H_t(s) = \frac{\theta(s)}{u_g(s) - u_d(s)}.$$

The measured response is approximated by a double integrator,

$$H_t(s) = \frac{K_t}{s^2},$$

which gives the time-domain step response

$$y(t) = K_t \frac{(t - t_0)^2}{2} a.$$

Using two sufficiently separated points, the gain is recovered from

$$K_t = \frac{2}{a} \left(\frac{\sqrt{y_2} - \sqrt{y_1}}{t_2 - t_1} \right)^2, \quad \boxed{K_t = 1.336}.$$

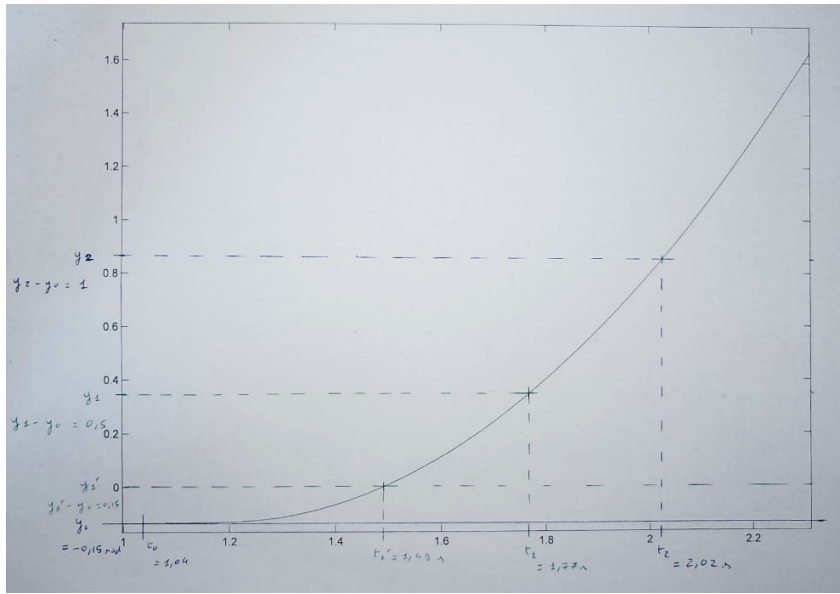


Figure 1.3: Experimental pitch step response used to identify the double-integrator gain.

1.4 Yaw-channel identification

DESIGN STAGE 4 Identification of pitch-to-yaw dynamics

Yaw is generated by the pitch inclination of the rotor cradle. The input is therefore θ and the output is λ ,

$$H_l(s) = \frac{\lambda(s)}{\theta(s)}.$$

The identified model contains an integrator and a first-order lag,

$$H_l(s) = \frac{K_l}{s(1 + \tau_l s)}.$$

For a step input a/s ,

$$\hat{y}(s) = \frac{K_l}{s(1 + \tau_l s)} \frac{a}{s} \implies y(t) = aK_l (t - \tau_l + \tau_l e^{-t/\tau_l}).$$

From the measured graph, $\tau_l = 20$ s and $a = -0.491$. With points $A(20, 0)$ and $B(60, 102.5)$,

$$K_I a = \frac{102.5 - 0}{60 - 20}, \quad \boxed{K_I = -5.219}.$$

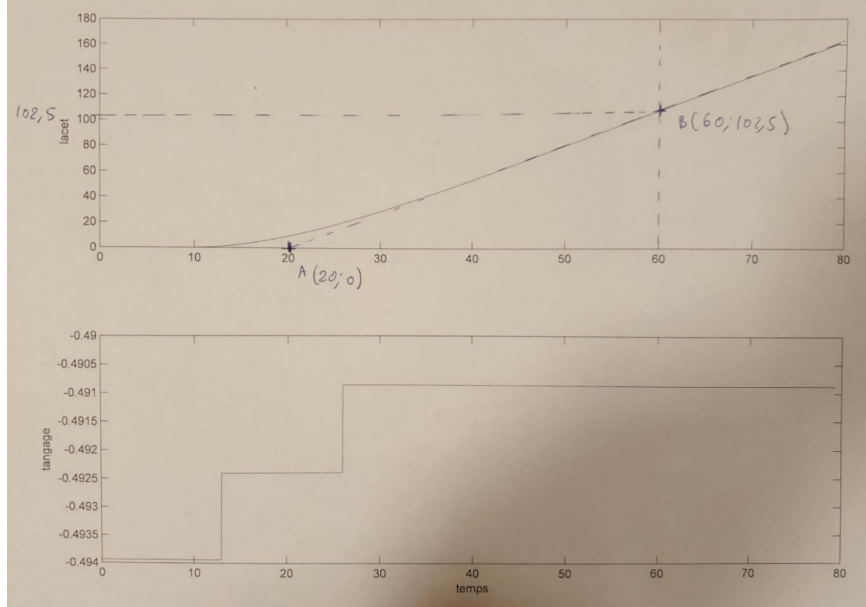


Figure 1.4: Yaw step-response measurement and input sequence.

1.5 Multivariable state-space model

DESIGN STAGE 5 Assembly of the full helicopter model

The three identified transfer functions lead to the differential equations

$$\ddot{\epsilon} = -2\xi\omega_0\dot{\epsilon} - \omega_0^2\epsilon + K_e\omega_0^2(U_g + U_d),$$

$$\ddot{\theta} = K_t(U_g - U_d),$$

$$\dot{v} = -\frac{1}{\tau_l}v + \frac{K_l}{\tau_l}\theta, \quad v = \dot{\lambda}.$$

The control input and regulated output vectors are

$$U = \begin{bmatrix} U_g \\ U_d \end{bmatrix}, \quad Y = \begin{bmatrix} \epsilon \\ v \end{bmatrix}.$$

A convenient state vector is

$$X = \begin{bmatrix} \dot{\epsilon} & \epsilon & \dot{\theta} & \theta & v \end{bmatrix}^T.$$

The model is

$$\dot{X} = AX + BU, \quad Y = CX,$$

with

$$A = \begin{bmatrix} -2\xi\omega_0 & -\omega_0^2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & K_t/\tau_l & -1/\tau_l \end{bmatrix},$$

$$B = \begin{bmatrix} K_e\omega_0^2 & K_e\omega_0^2 \\ 0 & 0 \\ K_t & -K_t \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Using the reported identified parameters,

$$A = \begin{bmatrix} -0.1120 & -0.4900 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -0.5219 & -0.1000 \end{bmatrix},$$

$$B = \begin{bmatrix} 0.1421 & 0.1421 \\ 0 & 0 \\ 1.3360 & -1.3360 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Modeling interpretation

The sum of the motor voltages controls elevation, whereas their difference controls pitch. Pitch acts as the intermediate variable that produces yaw-rate dynamics. The resulting plant is structurally multivariable even though the initial channel-identification experiments are performed independently.

Controller Synthesis and Comparative Evaluation

2.1 Integral state feedback by discrete pole placement

DESIGN STAGE 6 Refined elevation model for digital implementation

The pole-placement study uses a second elevation-identification campaign, focused specifically on the digital controller. Its measured response is shown in [figure 2.1](#).

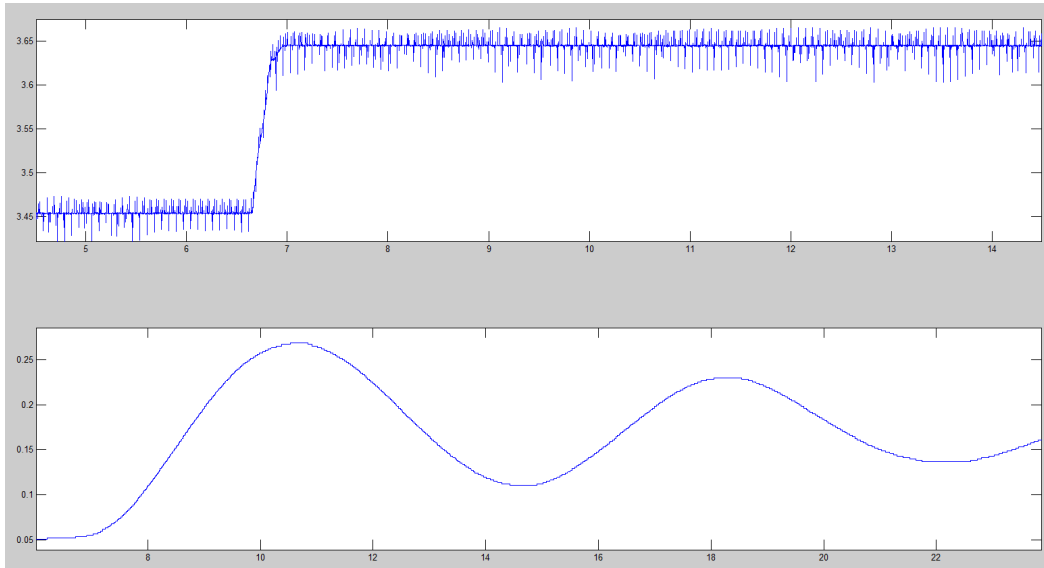


Figure 2.1: Refined elevation identification: applied voltage step and measured oscillatory response.

The reported elevation model is

$$H_e(s) = \frac{K}{s^2/\omega_0^2 + (2\xi/\omega_0)s + 1} = \frac{0.2131}{s^2 + 0.1465s + 0.6461},$$

with

$$\xi = 0.0912, \quad \omega_0 = 0.8038 \text{ rad s}^{-1}, \quad K = 0.3299 \text{ rad V}^{-1}.$$

The associated differential equation is

$$\ddot{\epsilon}(t) + 2\xi\omega_0\dot{\epsilon}(t) + \omega_0^2\epsilon(t) = \omega_0^2K u(t).$$

With $x = [\epsilon \ \dot{\epsilon}]^T$,

$$\dot{x} = Ax + Bu, \quad y = Cx + Du,$$

$$A = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & -2\xi\omega_0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \omega_0^2K \end{bmatrix}, \quad C = [1 \quad 0], \quad D = 0,$$

which numerically gives

$$A = \begin{bmatrix} 0 & 1 \\ -0.6461 & -0.146 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0.2652 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

2.1.1 Zero-order-hold discretization

For a sampling period $T_s = 0.05$ s, the zero-order-hold model is

$$x_{k+1} = Fx_k + Gu_k, \quad y_k = Cx_k,$$

where

$$F = e^{AT_s}, \quad G = \int_0^{T_s} e^{A\tau} B d\tau.$$

The identified discrete matrices are

$$F = \begin{bmatrix} 0.9992 & 0.0498 \\ -0.0322 & 0.9919 \end{bmatrix}, \quad G = \begin{bmatrix} 0.0003 \\ 0.0132 \end{bmatrix}.$$

2.1.2 Integral augmentation and controllability

The controller includes an integral state so that a constant reference is tracked without steady-state error. The augmented matrices are

$$F_e = \begin{bmatrix} F & 0 \\ C & 1 \end{bmatrix}, \quad G_e = \begin{bmatrix} G \\ 0 \end{bmatrix},$$

therefore

$$F_e = \begin{bmatrix} 0.9992 & 0.0498 & 0 \\ -0.0322 & 0.9919 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad G_e = \begin{bmatrix} 0.0003 \\ 0.0132 \\ 0 \end{bmatrix}.$$

The original discrete pair is controllable because

$$\det \begin{bmatrix} G & FG \end{bmatrix} = 0.0003(0.0131) - 0.0009(0.0132) = -7.9 \times 10^{-6} \neq 0.$$

The augmented matrix is nonsingular, and the additional eigenvalue at $z = 1$ represents the integral mode.

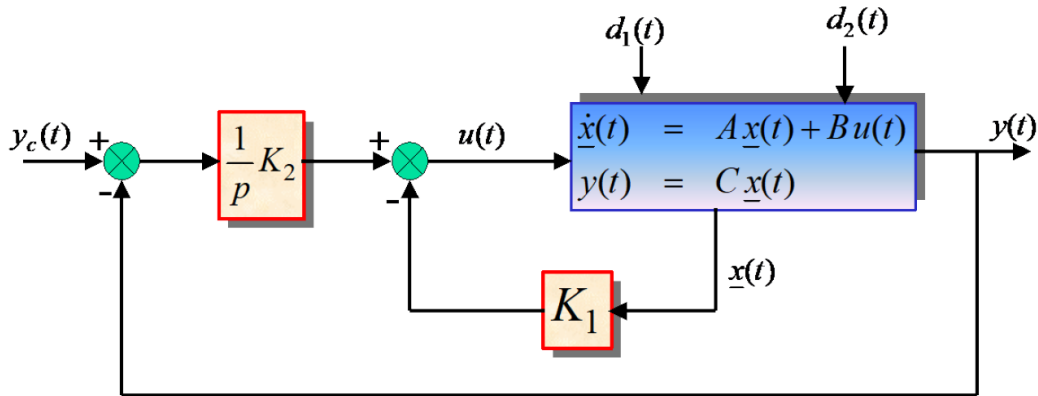


Figure 2.2: Integral state-feedback architecture with input and output disturbances.

2.1.3 Nominal pole placement

The sampling period is selected relative to the fastest desired closed-loop time constant,

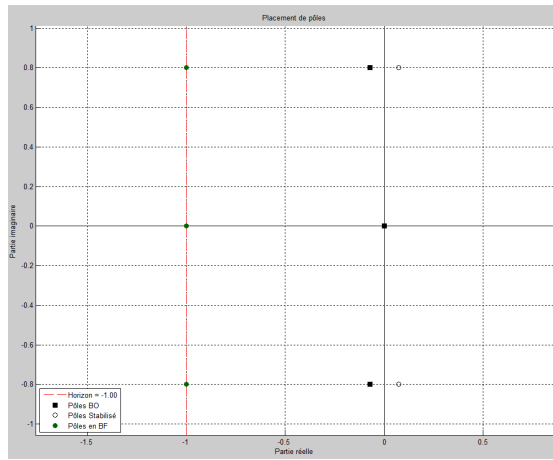
$$\frac{\tau_{\min}}{10} \leq T_s \leq \frac{\tau_{\min}}{5}.$$

The first design uses a continuous-time placement horizon at -1 , corresponding to $T_c = 1$ s. Mapping the desired poles into the z plane gives

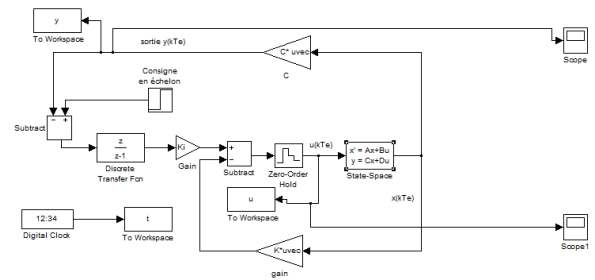
$$P_{BF,z} = \{0.9512, 0.9505 \pm 0.0381i\}.$$

The gain found by Ackermann pole placement is

$$K_p = \begin{bmatrix} 10.6434 & 10.2527 \end{bmatrix}, \quad K_i = 0.2882.$$



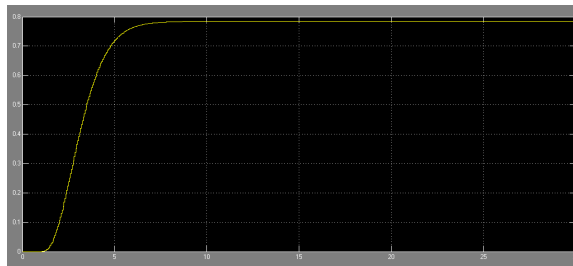
(a) Open-loop and nominal closed-loop poles.



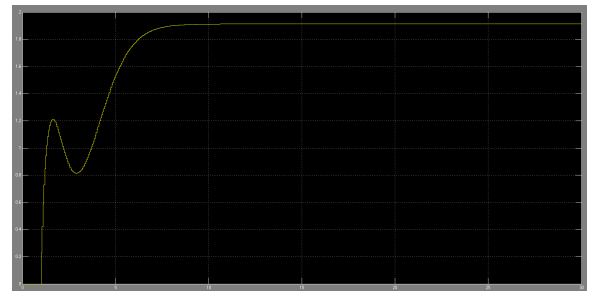
(b) Discrete integral-feedback Simulink model.

Figure 2.3: Nominal integral state-feedback design and simulation architecture.

The simulated response has zero steady-state error, no overshoot, and no sustained oscillation. The reported response time is approximately 6 s. The control voltage remains moderate, with a reported peak of approximately 1.9 V in the nominal simulation.



(a) Nominal elevation response.



(b) Nominal control command.

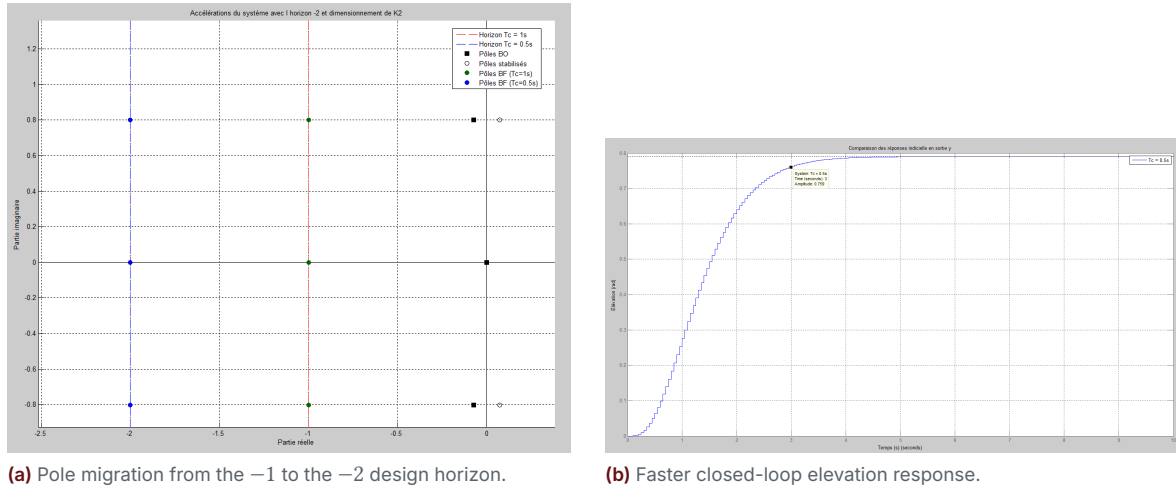
Figure 2.4: Closed-loop behavior with the nominal pole-placement controller.

2.1.4 Faster pole-placement design

To reduce the response time, the design horizon is moved from -1 to -2 , corresponding to $T_c = 0.5$ s. The resulting gain is

$$K_p = \begin{bmatrix} 40.3459 & 20.0418 \end{bmatrix}, \quad K_i = 1.5137.$$

The measured simulation response time decreases to approximately 3 s.



(a) Pole migration from the -1 to the -2 design horizon.

(b) Faster closed-loop elevation response.

Figure 2.5: Accelerated integral state-feedback design.

The faster response requires substantially greater actuator effort. In the comparative command plot, the $T_c = 0.5$ s design reaches approximately 3.3 V, whereas the $T_c = 1$ s design remains near 1.2 V. The aggressive design also exhibits a more pronounced transient. This experiment illustrates the central pole-placement trade-off: shifting poles farther left increases speed but also increases gain, control energy, noise sensitivity, and the risk of actuator saturation.

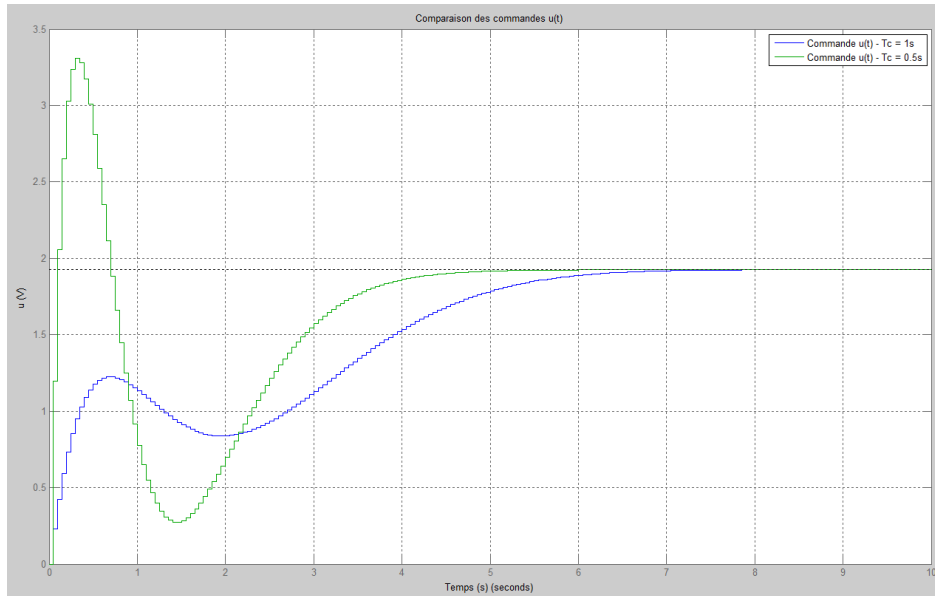


Figure 2.6: Comparison of control effort for the nominal and accelerated pole-placement designs.

2.1.5 Real-time implementation

The controller is deployed through MATLAB/Simulink and a dSPACE DS1103/1104 rapid-control-prototyping interface. The real-time architecture separates command generation, the digital control law, and the physical twin-rotor plant.

The physical platform stabilizes near the horizontal equilibrium. The measured response time is 5.78 s, close to the simulated value of 6.2 s. Manual perturbations are rejected in an average of 4.41 s. The small simulation-to-experiment discrepancy is consistent with unmodeled nonlinearities, actuator dynamics, sensor noise, and interface delays.

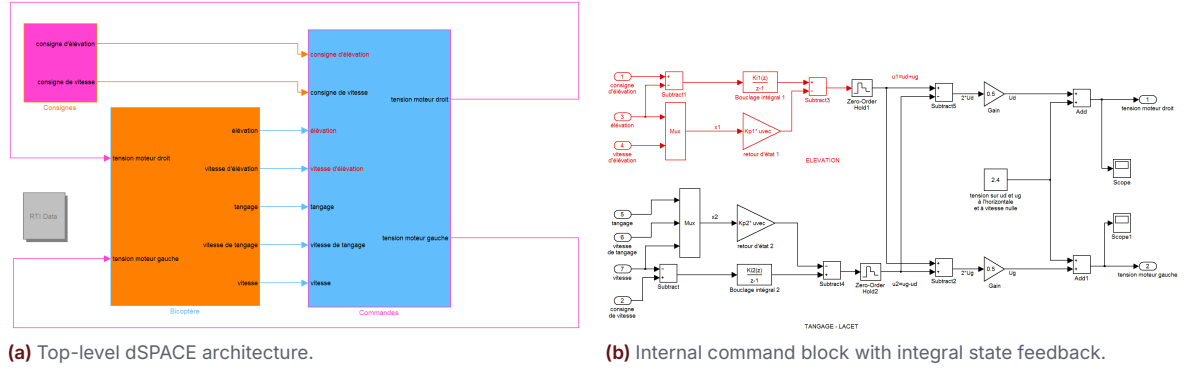


Figure 2.7: Real-time implementation of the pole-placement controller.

2.2 Linear-quadratic regulation with integral action

DESIGN STAGE 7 Multivariable optimal-control synthesis

The LQR design uses the five-state plant from Chapter 1 and augments it with two integral states, one for elevation and one for yaw rate. With $n = 5$ plant states and $p = 2$ regulated outputs,

$$A_a = \begin{bmatrix} A & 0_{n \times p} \\ C & 0_{p \times p} \end{bmatrix}, \quad B_a = \begin{bmatrix} B \\ 0_{p \times m} \end{bmatrix}.$$

Numerically,

$$A_a = \begin{pmatrix} -0.1120 & -0.4900 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.5219 & -0.1000 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix},$$

$$B_a = \begin{pmatrix} 0.1421 & 0.1421 \\ 0 & 0 \\ 1.3360 & -1.3360 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

2.2.1 Quadratic criterion and spectral shift

The control law minimizes

$$J = \int_0^\infty (x^T Q x + u^T R u) dt,$$

with

$$Q = \text{diag}(0, 0, 0, 0, 0, 40, 40), \quad R = I_2.$$

Only the integrated tracking errors are directly penalized. The two motor commands are weighted equally. A spectral shift $\alpha_c = 0.5$ is applied,

$$A_{a,\alpha} = A_a + \alpha_c I_7.$$

The algebraic Riccati equation is

$$PA_a + A_a^T P - PB_a R^{-1} B_a^T P + Q = 0.$$

The reported Riccati solution is

$$P = \begin{pmatrix} 87.1427 & 119.5267 & 0 & 0 & 0 & 79.0374 & 0 \\ 119.5267 & 278.1660 & 0 & 0 & 0 & 207.9665 & 0 \\ 0 & 0 & 1.7422 & 4.5466 & -11.5468 & 0 & -7.6855 \\ 0 & 0 & 4.5466 & 17.7038 & -53.7353 & 0 & -40.1133 \\ 0 & 0 & -11.5468 & -53.7353 & 189.5634 & 0 & 162.1503 \\ 79.0374 & 207.9665 & 0 & 0 & 0 & 212.2804 & 0 \\ 0 & 0 & -7.6855 & -40.1133 & 162.1503 & 0 & 170.8591 \end{pmatrix}.$$

The optimal gain $K = R^{-1} B_a^T P = [K_p \ K_i]$ is

$$K_p = \begin{pmatrix} 12.3830 & 16.9847 & 2.3276 & 6.0742 & -15.4265 \\ 12.3830 & 16.9847 & -2.3276 & -6.0742 & 15.4265 \end{pmatrix},$$

$$K_i = \begin{pmatrix} 11.2312 & -10.2679 \\ 11.2312 & 10.2679 \end{pmatrix}.$$

The closed-loop eigenvalues are

$$\{-0.5607 \pm 0.9945i, -0.6338 \pm 1.3929i, -1.0098, -1.5259 \pm 0.5786i\},$$

so every pole has a negative real part.

2.2.2 Nominal and disturbed simulations

The nominal simulation regulates elevation and yaw rate simultaneously. The reported 5% response times are approximately 3.1 s for elevation and 2.7 s for yaw rate. No overshoot is observed, and the two motor commands remain below their saturation limits.

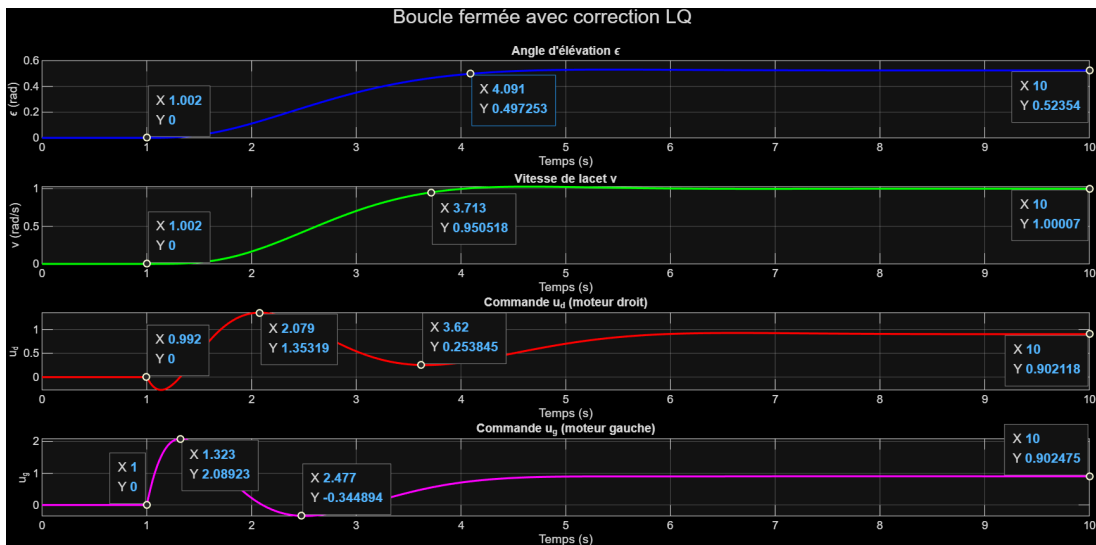


Figure 2.8: Nominal multivariable closed-loop response with the LQR controller.

Increasing Q would accelerate tracking at the cost of higher control energy; increasing R would reduce

actuator demand at the cost of slower tracking. The selected weights produce a balanced response without visible saturation. To evaluate robustness, simultaneous two-second rectangular disturbances are applied to the symmetric elevation channel and the differential yaw channel.

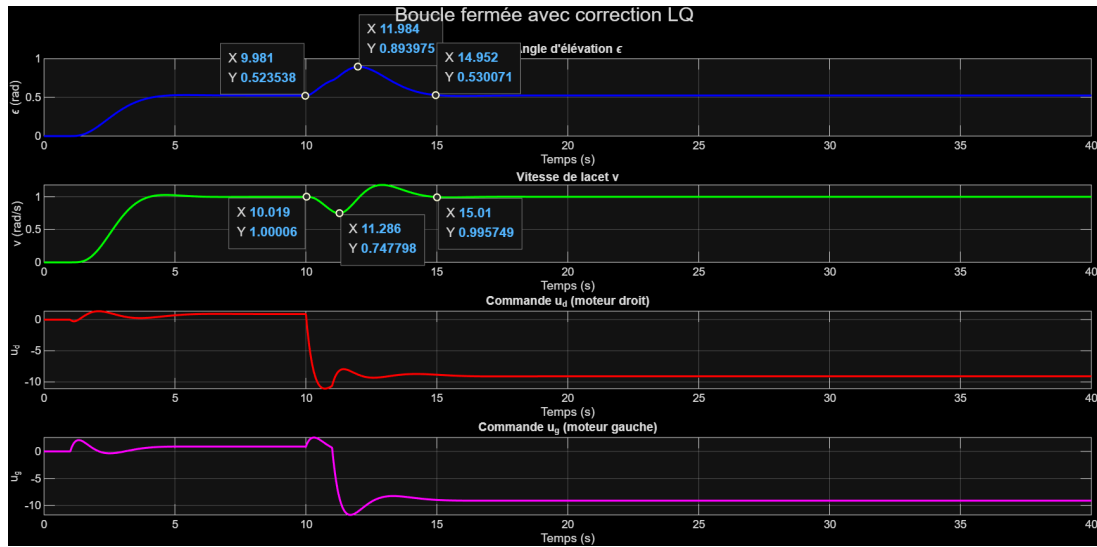


Figure 2.9: LQR response to simultaneous elevation and yaw-channel disturbances.

Both regulated outputs return to their references in approximately 5 s without sustained oscillations. Real-time dSPACE tests reproduce the qualitative simulation behavior: the platform is stabilized, reference tracking is maintained, and manually applied disturbances are rejected within a few seconds.

2.3 Performance comparison

DESIGN STAGE 8 Engineering comparison of the two control strategies

The two controller families address related but not identical problems. Integral pole placement is developed for the elevation channel and gives direct control over the desired closed-loop poles. LQR is developed for the full two-input, two-output model and determines the gain from a global optimization criterion.

Table 2.1: Comparison of integral pole placement and integral LQR.

Criterion	Integral state feedback	Integral LQR
Controlled scope	SISO elevation channel	MIMO elevation and yaw-rate channels
Primary tuning variables	Desired pole locations / T_c	Weighting matrices Q , R , and spectral shift α_c
Steady-state error	Zero through one integral state	Zero through two integral states
Nominal reported response	About 6 s for $T_c = 1$ s; about 3 s for $T_c = 0.5$ s	About 3.1 s elevation and 2.7 s yaw rate
Control effort	Explicitly rises as poles move left; about 3.3 V for the faster design	Limited systematically through R ; no simulated saturation
Disturbance rejection	About 4.41 s in real manual tests	About 5 s for simultaneous simulated channel disturbances
Design transparency	Direct modal interpretation and straightforward implementation	Systematic optimal compromise, naturally multivariable
Main limitation	Gains become aggressive as response speed is increased	Performance depends on physically meaningful weight selection and model fidelity

Comparative interpretation

The pole-placement design is particularly effective when a dominant channel can be isolated and a specific transient envelope is required. Its main advantage is directness: moving the poles gives a predictable change in response speed. Its main drawback is equally direct: faster poles require larger gains and greater actuator effort.

The LQR design is more suitable for the coupled helicopter because it treats both motor commands and both regulated outputs in one synthesis. Rather than selecting every pole explicitly, the designer specifies the relative importance of tracking error and control energy. The resulting controller is therefore easier to scale toward coordinated multivariable operation, although the quality of the result depends on the state model and the interpretation of Q and R .

Conclusion

This project combined two control studies into a single investigation of a twin-rotor helicopter platform. Open-loop experiments first established complementary models for elevation, pitch, and yaw. Elevation was represented by a lightly damped second-order mode, pitch by a double integrator, and yaw by an integrating first-order model. These channel descriptions were then assembled into a five-state, two-input, two-output representation suitable for modern state-space control.

The discrete integral state-feedback study demonstrated that pole placement can transform the unstable or poorly damped elevation dynamics into a smooth response with zero steady-state error. The nominal design produced a response time close to 6 s in both simulation and experiment, while rejecting manual disturbances in approximately 4.41 s. Moving the design horizon from -1 to -2 reduced the simulated response time to approximately 3 s, but the associated command peak rose to roughly 3.3 V. The experiment therefore made the speed-effort trade-off explicit.

The LQR study extended the control problem to simultaneous elevation and yaw-rate regulation. Integral augmentation guaranteed steady-state accuracy, while the Riccati-based synthesis provided a systematic compromise between accumulated tracking error and motor effort. The reported responses were fast, non-oscillatory, and free from actuator saturation, with settling times of approximately 3.1 s and 2.7 s for elevation and yaw rate. Simultaneous disturbances were rejected in approximately 5 s, and real-time dSPACE tests confirmed stable physical operation.

Taken together, the results show that neither approach is universally superior. Integral pole placement is preferable when direct pole geometry, transparent transient shaping, and a focused SISO implementation are the dominant requirements. Integral LQR is preferable when several coupled states and actuator channels must be coordinated and when actuator effort must be included directly in the design objective. For the complete helicopter platform, the multivariable LQR provides the stronger foundation for future development.

The next logical extensions are state estimation through a Luenberger observer or Kalman filter, explicit treatment of measurement noise and parameter uncertainty, inclusion of the pitch channel in the regulated output vector, and evaluation of robust or predictive methods such as H_∞ control and model predictive control. These developments would move the platform from a linear laboratory benchmark toward a more complete architecture for uncertain, constrained, and nonlinear aerial systems.